

CHARGING LITHIUM-ION BATTERIES

Properly restoring energy using the latest battery technology for today's portable products requires careful consideration. An understanding of the charging characteristics of the battery and the application's requirements is essential in order to design an appropriate and reliable battery charging system.

Linear and switch-mode charging solutions for Li-ion batteries are presented here. Also given are the guidelines and considerations that should be taken into account when developing any battery charging system



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Powering today's portable world poses many challenges for system designers. The use of

batteries as a prime power source is on the rise. As a result, a burden has been placed on the system designer to create sophisticated systems utilising the battery's full potential.

Each application is unique, but one common theme rings through: maximise battery capacity usage. This theme directly relates to how energy is properly restored to rechargeable batteries. No single method is ideal for all applications. An understanding of the charging characteristics of the battery and the application's requirements is essential in order to design an appropriate and reliable battery charging system. Each method has its associated advantages and disadvantages.

It is the particular application with its individual requirements that determines which method will be the best to use.

Far too often, the charging system is given low priority, especially in cost-sensitive applications. The quality of the charging system, however, plays a key role in the life and reliability of the battery. In this article, the fundamentals of charging Lithium-ion (Li-ion) batteries are explored. In particular, linear charging solutions and a micro-controller-based, switch-mode solution shall be explored.

Li-ion charging

The rate of charge or discharge is often expressed in relation to the capacity of the battery. This rate is known as the C-rate. The C-rate equates to a charge or discharge current and is defined as:

$$I = M \times C_n$$

where 'I' is charge or discharge current